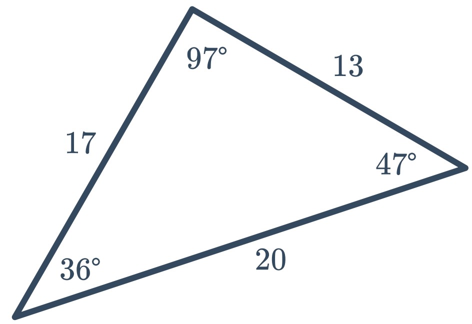
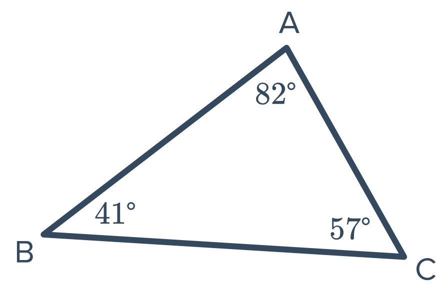
Inequalities in one triangle

1 Consider the given triangle and think about the longest and shortest sides.

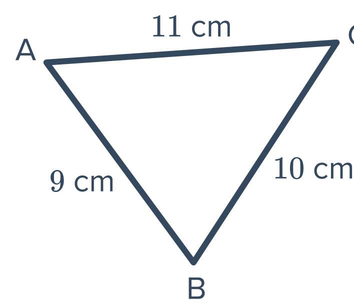


- **a** Find the sum of the interior angles in the triangle.
- **b** Which side appears opposite the largest angle?
- **c** Which side appears opposite the smallest angle?

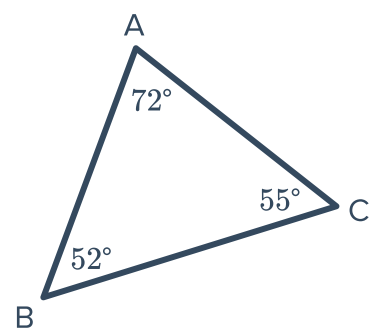
2 Consider the given triangle.
Which side is the longest side of the triangle?



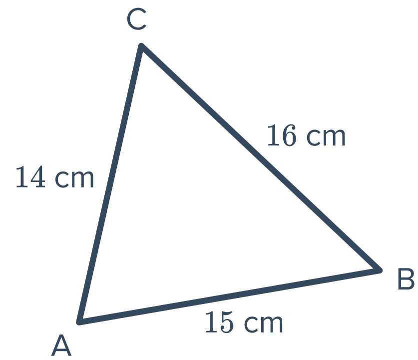
3 Consider the given triangle.
Which angle is the largest angle of the triangle?



4 Consider the given triangle.
Which side is the shortest side of the triangle?



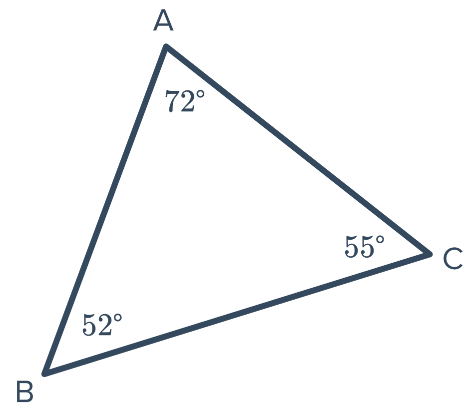
5 Consider the given triangle.
Which angle is the smallest angle of the triangle?



6 Consider the given triangle.

a List the sides of the triangle from smallest to largest.

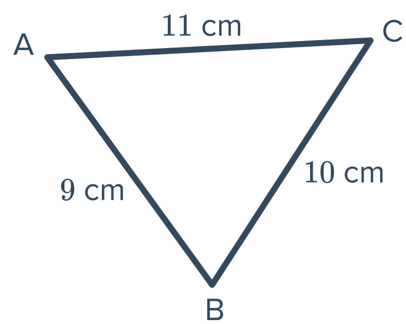
b Write an inequality relating the lengths of the three sides of the triangle.



7 Consider the given triangle.

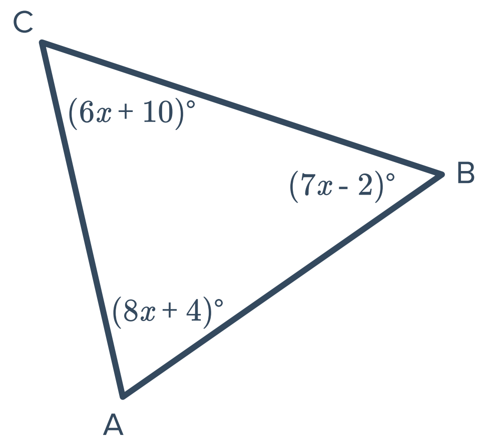
a List the angles of the triangle from smallest to largest.

b Write an inequality relating the measure of the three angles of the triangle.



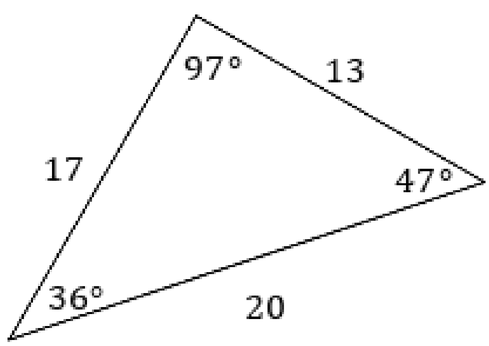
8 Consider the given triangle.

- **a** Solve for x.
- **b** Find $m\angle CAB$.
- **c** Find $m\angle ABC$.
- **d** Find $m\angle BCA$.
- **e** Hence, which side is the longest side of the triangle?



Additional questions

9 Consider the triangle in the given figure. Describe the relationship between the sizes of the angle and opposite side compared to the other opposite side-angle pairs.

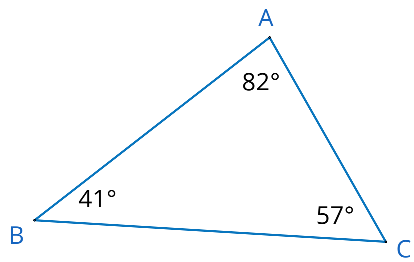


10 Complete the following theorems:

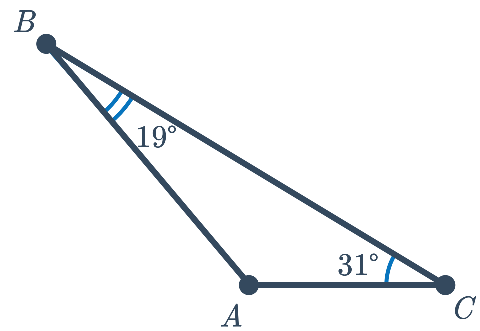
- **a** If one side of a triangle is longer than another side, then the angle opposite the longer side is...
- **b** If one angle of a triangle is larger than another angle, then the side opposite the larger angle is...

11 For each of the following triangles, order th[redacted]s from shortest to longest.

a



b



c $\triangle ABC$ where,

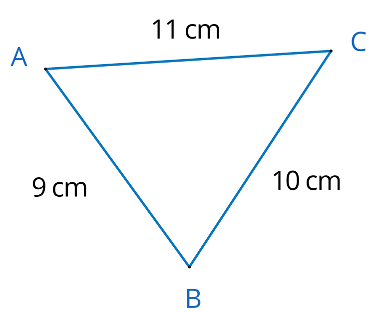
- $m\angle A = 37^{\circ}$
- $m\angle B = 90^{\circ}$
- $m\angle C = 53^{\circ}$

d $\triangle ABC$ where,

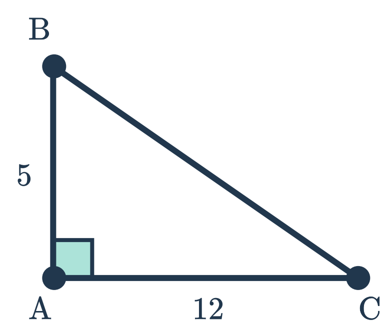
- $m\angle A = 115^{\circ}$
- $m\angle B = 32^{\circ}$

12 For each of the following triangles, order the angles from smallest to largest.

a



b



c $\triangle ABC$ where,

- $AB = 7$
- $AC = 3.2$
- $BC = 4.8$

d $\triangle ABC$ where,

- $AB = 50$
- $AC = 62$
- $BC = 70$

13 Draw the following triangles using the give[redacted]sures.

a Side lengths

- 12
- 6
- 8

Angle measures

- 24°
- 120°
- 36°

b Side lengths

- 8
- 6
- 10

Angle measures

- 53°
- 37°

14 Given that the hypotenuse is the longest side in in a right triangle, explain why a right triangle must have one right angle and two acute angles, without using the triangle angle sum theorem.